



▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

Executive Summary

Fairness Experiment Report - Upload 28 summarizes the current fairness state for upload 28. Accuracy changed to +0.782 and fairness score changed to +0.954. Strong fairness improvement with no measurable accuracy loss. The best balanced model is ThresholdOptimizer_mitigated_threshold with combined score 0.885.

ACCURACY	FAIRNESS SCORE	COMBINED SCORE	DISPARATE IMPACT
0.782 ▲ +0.782	0.954 ▲ +0.954	0.885 ThresholdOptimizer_mitigated_threshold	0.774 Ratio (DIR)

Dataset & Model Information

DATASET	MODEL	SECTIONS INCLUDED
processed_dataset.csv	compas_dt_baseline.pkl	comparison

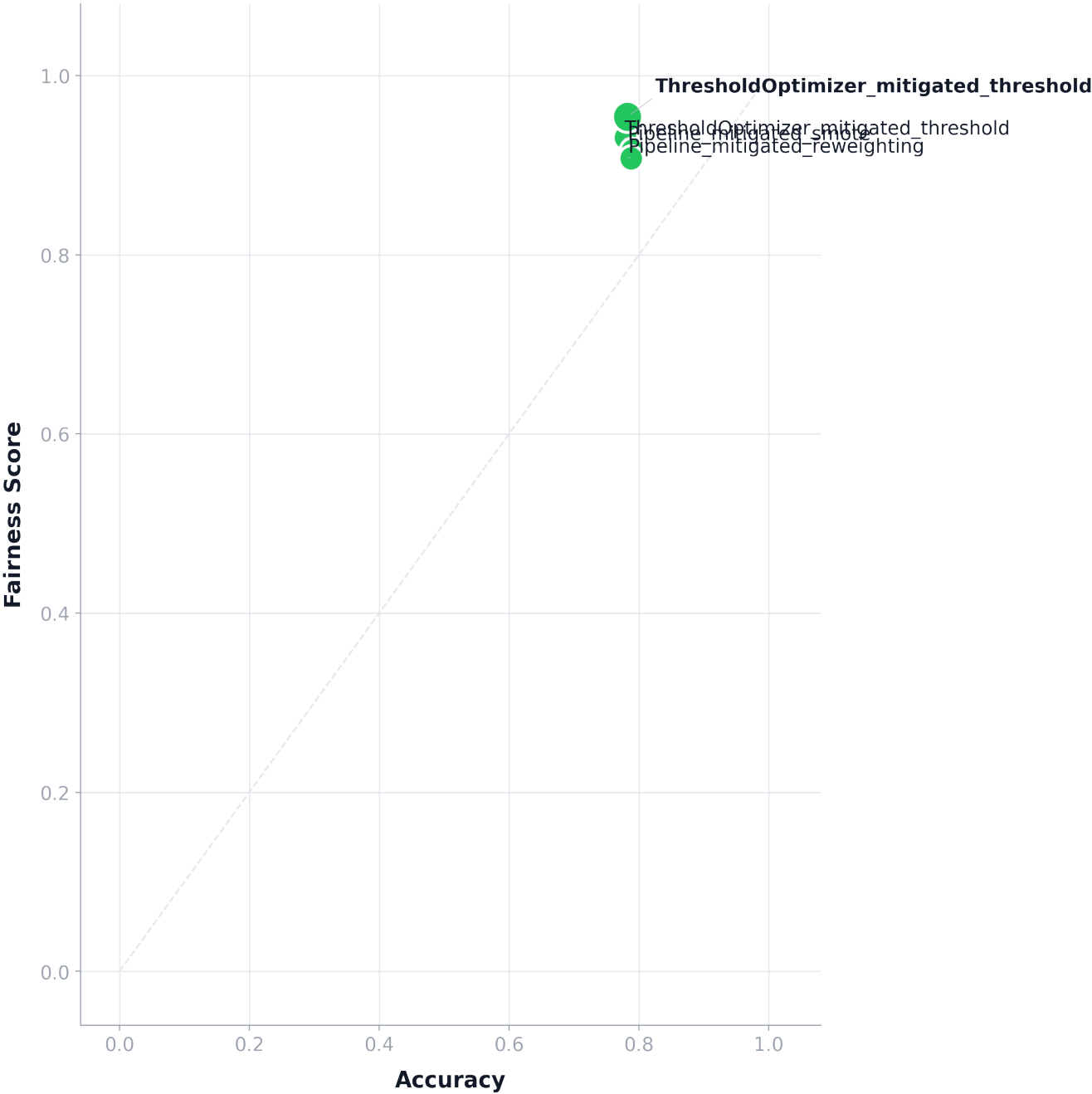
Fairness Metrics

Metric	Value	Interpretation
Demographic Parity Difference (DPD)	0.092	Difference in positive-prediction rates across groups; closer to 0 is more equitable.
Equalized Odds Difference (EOD)	0.000	Gap in true/false positive rates between groups; closer to 0 indicates equal error rates.
Disparate Impact Ratio (DIR)	0.774	Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.

Model Comparison

Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ ThresholdOptimizer_mitigated_threshold	0.782	0.954	0.092	0.000	0.774	0.885
★ ThresholdOptimizer_mitigated_threshold	0.779	0.931	0.103	0.035	0.747	0.870
Pipeline_mitigated_smote	0.789	0.916	0.139	0.028	0.672	0.866
Pipeline_mitigated_reweighting	0.788	0.908	0.144	0.040	0.654	0.860

Accuracy vs. Fairness Trade-off



Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.